

A Factorization Identity for Twisted Multinomial Coefficients with Application to Pilot States in Hamiltonian Decoded Quantum Interferometry

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Based on

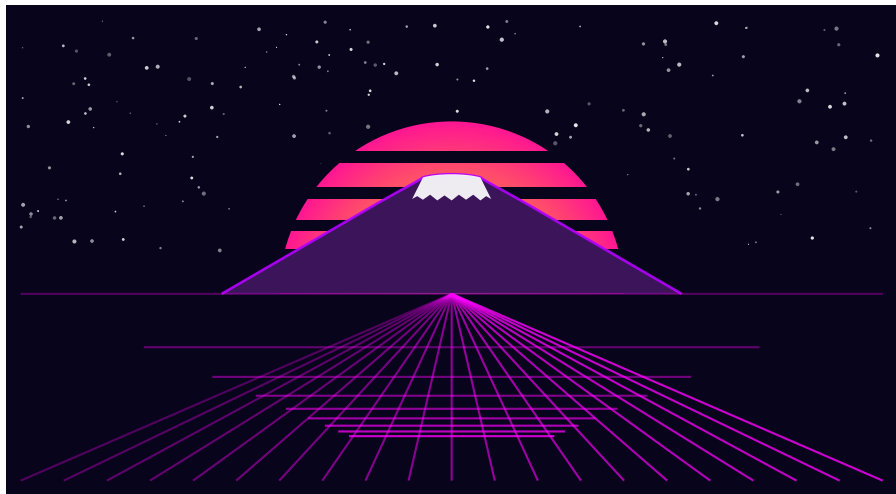
A Factorization Identity for Twisted Multinomial Coefficients with Application to Pilot States in Hamiltonian Decoded Quantum Interferometry

Paweł Wocjan, arXiv:2604.01022 (April 2026)

Builds on:

- ▶ A. Schmidhuber, J. Lu, N. Shutty, S. Jordan, A. Poremba, Y. Quek, *Hamiltonian Decoded Quantum Interferometry*, arXiv:2510.07913.
- ▶ K. Bu, W. Gu, X. Li, *Hamiltonian decoded quantum interferometry for general Pauli Hamiltonians*, arXiv:2601.18773.

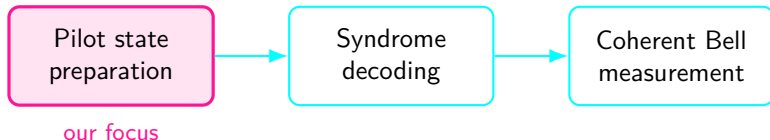
Enkei – Distant View



Enkei Summary I – HDQI

- ▶ **Hamiltonian Decoded Quantum Interferometry**
(Schmidhuber et al. 2025): a quantum algorithm that prepares Gibbs and ground states of many-body Hamiltonians.
- ▶ Inside HDQI, the central data is a so-called **pilot state**.

HDQI – the protocol in three steps



- ▶ We address only the *first* step.
- ▶ Decoding is a separate problem; both must work together for HDQI to succeed.

Enkei Summary II – the bottleneck

- For an n -qubit Pauli Hamiltonian

$$H = \sum_{j=1}^m c_j P_j,$$

the pilot state encodes the coefficients of H^k in the Pauli basis:

$$H^k = \sum_{r \in \{0,1\}^m} \alpha_r P_1^{r_1} \cdots P_m^{r_m}.$$

- There are 2^m amplitudes – exponentially many.
- *Computing them efficiently is the bottleneck.*

Enkei Summary III – our contribution

- ▶ We identify a structural condition on the Pauli commutation pattern – **predecessor-uniformity** – that yields:
 - ▶ a clean **combinatorial identity** for so-called *twisted multinomial coefficients*, generalizing the classical q -multinomial,
 - ▶ an **exact MPS of bond dimension** $k + 1$ for the pilot state amplitudes.
- ▶ For mutually anticommuting generators, this is an *exponential improvement* over previous methods.

Chūkei – Middle View



A familiar object – the multinomial coefficient

- ▶ $\binom{k}{k_1, \dots, k_m}$ counts arrangements of the multiset $\{1^{k_1}, \dots, m^{k_m}\}$.
- ▶ *Example.* $\binom{3}{2,1} = 3$ arrangements of two 1's and one 2:

1 1 2 1 2 1 2 1 1

Refining by inversions – the q -multinomial

- ▶ Weight each arrangement by $q^{\# \text{inversions}}$, where an *inversion* is a pair of positions where a larger letter precedes a smaller one.
- ▶ *Same example.*

$$1\ 1\ 2 \rightarrow 0 \text{ inversions} \rightarrow 1$$

$$1\ 2\ 1 \rightarrow 1 \text{ inversion} \rightarrow q$$

$$2\ 1\ 1 \rightarrow 2 \text{ inversions} \rightarrow q^2$$

- ▶ Sum: $\binom{3}{2,1}_q = 1 + q + q^2$.

The classical factorization

- The q -multinomial factors as a product of *Gaussian binomials*:

$$\binom{k}{k_1, \dots, k_m}_q = \prod_{j=1}^m \binom{\ell_j}{k_j}_q, \quad \ell_j = k_1 + \dots + k_j.$$

- Picture: build any arrangement **one letter type at a time**, interleaving the k_j copies of letter j into the existing word.
Each stage contributes one Gaussian binomial.

The new object – twisted multinomial

- ▶ In the q -multinomial, every inversion has the *same* weight q .
- ▶ **Twisted version.** Each inversion of types $i < j$ carries its own weight ω_{ij} :

$$\binom{k}{k_1, \dots, k_m}_{\Omega} := \sum_{\sigma} \prod_{\substack{\text{inversions} \\ \text{of type } (i,j)}} \omega_{ij}.$$

- ▶ Specializing $\omega_{ij} \equiv q$ recovers the q -multinomial.
- ▶ In general: *no clean factorization*.

Concrete example – $\binom{3}{1,1,1}_\Omega$

Six arrangements of $\{1, 2, 3\}$, each with its inversion weight:

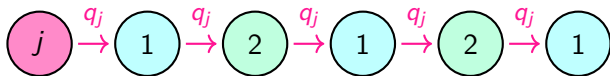
arrangement	weight
1 2 3	1
1 3 2	$\omega_{2,3}$
2 1 3	$\omega_{1,2}$
2 3 1	$\omega_{1,2} \omega_{1,3}$
3 1 2	$\omega_{1,3} \omega_{2,3}$
3 2 1	$\omega_{1,2} \omega_{1,3} \omega_{2,3}$

$$\binom{3}{1,1,1}_\Omega = 1 + \omega_{1,2} + \omega_{2,3} + \omega_{1,2}\omega_{1,3} + \omega_{1,3}\omega_{2,3} + \omega_{1,2}\omega_{1,3}\omega_{2,3}.$$

What makes the twisted multinomial factor?

- ▶ In the classical proof, all stages of the build-up contribute the *same* weight q , so they decouple.
- ▶ In the twisted case, the weight of inserting a letter j depends on *which* earlier letters it crosses.
- ▶ We need a structural condition to make stages decouple again.
- ▶ Enter *predecessor-uniformity*.

Predecessor-uniformity – the picture



Predecessor-uniform: when letter j crosses any earlier letter $i < j$, the cost is the *same* q_j – regardless of which earlier letter type it is.

Formally: $\omega_{ij} = q_j$ for all $i < j$.

Whether Ω admits a predecessor-uniform ordering is decidable greedily in $O(m^3)$.

Main theorem

Proposition. *If Ω is predecessor-uniform with parameters $q_2, q_3, \dots, q_m$, then*

$$\binom{k}{k_1, \dots, k_m}_{\Omega} = \prod_{j=1}^m \binom{\ell_j}{k_j}_{q_j}, \quad \ell_j = k_1 + \dots + k_j.$$

- ▶ Same shape as the classical product formula.
- ▶ One q becomes $m-1$ independent parameters q_j – each Gaussian binomial uses its own q_j .
- ▶ Holds for arbitrary $q_j \in \mathbb{C} \setminus \{0\}$ – no algebraic constraint required.

Verifying on the example

For $\binom{3}{1,1,1}_\Omega$ under predecessor-uniformity:

$$\omega_{1,2} = q_2, \quad \omega_{1,3} = \omega_{2,3} = q_3$$

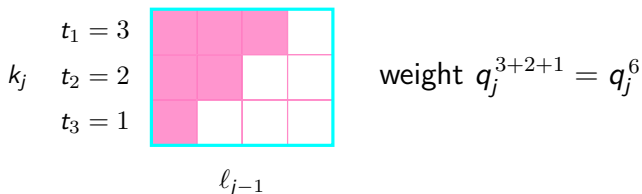
Substituting into each arrangement's weight:

arrangement weight		arrangement weight	
1 2 3	1	2 3 1	$q_2 q_3$
1 3 2	q_3	3 1 2	q_3^2
2 1 3	q_2	3 2 1	$q_2 q_3^2$

$$\begin{aligned}\binom{3}{1,1,1}_\Omega &= 1 + q_2 + q_3 + q_2 q_3 + q_3^2 + q_2 q_3^2 \\ &= (1 + q_2)(1 + q_3 + q_3^2) = \binom{2}{1}_{q_2} \binom{3}{1}_{q_3} \quad \checkmark\end{aligned}$$

The proof in one picture

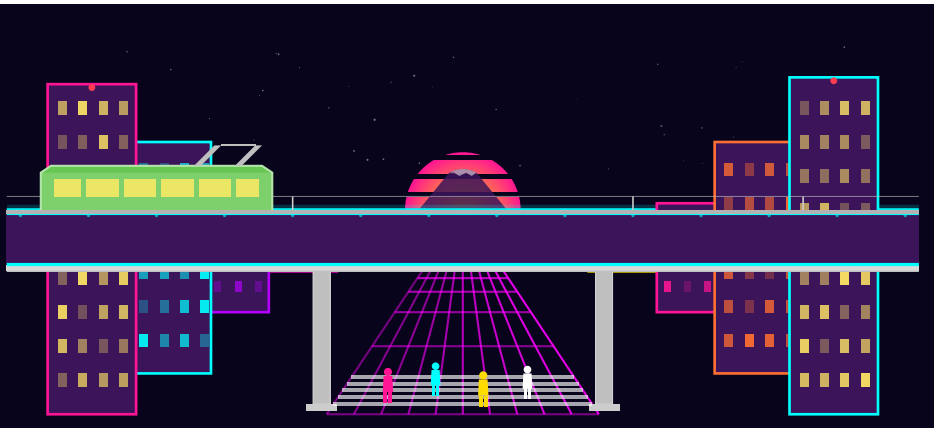
At stage j , the u -th inserted copy of j skips t_u earlier letters. The t_u 's form a partition λ in a $k_j \times \ell_{j-1}$ box; each skip contributes q_j .



► Sum over λ :
$$\sum_{\lambda \subseteq k_j \times \ell_{j-1}} q_j^{|\lambda|} = \binom{\ell_j}{k_j}_{q_j}.$$

► Predecessor-uniformity decouples the stages $\Rightarrow$ product formula. □

Kinkei – Close-up View



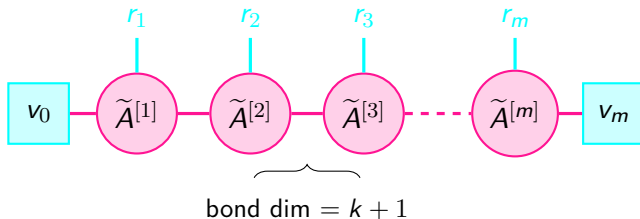
Where the twisted multinomial enters

- ▶ Pauli generators commute up to a sign:

$$P_i P_j = \omega_{ij} P_j P_i, \quad \omega_{ij} \in \{+1, -1\}.$$

- ▶ Each pilot-state amplitude α_r is a sum of **twisted multinomial coefficients** (with this Ω), weighted by products of Hamiltonian coefficients c_j .
- ▶ *Computing pilot states $\iff$ evaluating sums of twisted multinomials.*

From factorization to MPS



- Compact representation of the *coefficient tensor* α_r .
- Each $\tilde{A}_{r_j}^{[j]} \in \mathbb{C}^{(k+1) \times (k+1)}$ is built from a single Gaussian binomial $\binom{\ell_j}{\ell_j - \ell_{j-1}}_{q_j}$.
- $\alpha_r = v_0 (\prod_j \tilde{A}_{r_j}^{[j]}) v_m^\top$.

Why bond dimension $k + 1$ matters

- ▶ Bond dimension $k + 1$ is *independent of m* .
- ▶ Standard MPS preparation $\Rightarrow$ pilot state preparable in $\text{poly}(m, k)$ gates.
- ▶ For any polynomial $\mathcal{P}(H)$ of degree $\leq d$: *same* site matrices, bond dim $d + 1$ – only the right boundary vector changes.

Exponential separation – anticommuting Paulis

- ▶ Take $\{P_1, \dots, P_m\}$ pairwise anticommuting (e.g. Majoranas).
Then Ω is predecessor-uniform under *any* ordering ($q_j = -1$).
- ▶ Previous methods factor along connected components of the *anticommutation graph* – here the complete graph K_m .
Cost: $\exp(m)$.
- ▶ Our MPS: cost $O(mk^2)$ per amplitude.
Concretely, $m = 20$, $k = 4$: $\sim 10^6$ vs. ~ 320 .

Limitation – predecessor-uniformity vs. locality

- ▶ In the Pauli case: if P_j anticommutes with even one earlier generator, predecessor-uniformity forces P_j to anticommute with *all* earlier generators.

$\Rightarrow P_j$ shares support with *every* earlier generator.

- ▶ Tension with geometric locality: lattice generators with disjoint support always commute.
- ▶ Predecessor-uniform Hamiltonians are *highly non-local*.
Mutually-anticommuting Majoranas are the prototype.

Summary

- ▶ Introduced the **twisted multinomial coefficient** – pair-dependent inversions.
- ▶ Identified **predecessor-uniformity** as the structural condition that makes it factor into Gaussian binomials with *site-dependent* parameters.
- ▶ This gives an **exact MPS of bond dimension** $k + 1$ for HDQI pilot-state amplitudes – exponentially better than previous methods on mutually-anticommuting families.
- ▶ HDQI also requires efficient *decoding*: finding Hamiltonians where both conditions hold simultaneously is the central open problem.

Open problems

- ▶ **Combinatorial side.** Complexity of evaluating $\binom{k}{k_1, \dots, k_m}_\Omega$ for general (non-predecessor-uniform) Ω ?
- ▶ **Quantum side.** Hamiltonians admitting both efficient pilot-state preparation *and* efficient Hamiltonian-code decoding?
- ▶ **Generalizations.** Block-structured analogues (predecessor-uniform within blocks) might lift the locality obstruction.

